GGG-D-651d
October 21, 1964

SUPERSEDING
Int. Fed. Spec. GGG-D-00651c
January 10, 1963 and
Fed. Spec. GGG-D-651b
July 13, 1961

FEDERAL SPECIFICATION

DRILL, BREAST

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers breast drills generally used in hand drilling of wood and metal.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Type and styles.* Breast drills shall be two-speed, of one type and furnished in the following styles as specified, (see 6.2):

Style B—Universal two-jaw chuck.
Style C—Three-jaw chuck.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposals, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washinong, D. C., Atlanta, Chicago, Kansas City, Mo. Dallas Denver San Francisco Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5110

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3. REQUIREMENTS

3.1 Material. Breast drills shall be of malleable iron, steel, or other metal of such quality as to produce drills conforming to the requirements as hereinafter specified, and to withstand without injury the maximum strain to which they may be subjected under ordinary service conditions.

3.2 Design. Breast drills shall be of two-speed design and shall be properly proportioned in all parts so as to be strong, durable, and easy to operate. The breast drills shall have an overall length of not less than 15 nor more than 19 inches, with a maximum weight of 8 pounds.

3.2.1 *Illustration.* The illustration shown herein is for the convenience of requisitioning and purchasing officers, and is not intended to preclude the purchase of breast drills which are otherwise in accordance with the requirements of this specification.

3.3 Repair parts. The contractors supplying breast drills covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances and clearances, in order that such parts may be interchangeable, replaced, or adjusted without requiring modification.

3.4 Identification marking. Breast drills shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.5 Component parts. The breast drills shall be composed essentially of a frame, spindle, gears, speed-changing mechanism, chuck, crank arm, breastplate, and wooden handles.

3.5.1 *Frame.* The frame shall be of malleable or gray iron, or other metal suitable for the purpose intended. Frames shall be of strong, durable construction and capable of meeting the test specified in 4.3.1. All bearing surfaces shall have a smooth machine finish and shall be free from any imperfections which would adversely affect the serviceability.

3.5.2 *Spindle.* The spindle shall be of high-grade steel, of either integral or two-piece design. The chuck end of the spindle shall be suitably threaded to properly accommodate the chuck for a distance not less than the spindle diameter. The upper portion of the spindle shall be provided with a bevel driving pinion or pinions. When specified, (see 6.2), an idler gear shall be furnished. The spindle shall have ball or roller bearings to receive any thrust load applied during drill operations. The desired clearance between the bearings and spindle should not exceed 0.005 inch, but in any event shall be not more than 0.008 inch. Spindle-end play shall not exceed 0.010 inch. The parts shall be assembled in a manner to preclude the likelihood of their becoming loose under normal service conditions.

3.5.3 *Gears.* All gears shall have machine-cut or uniformly formed beveled teeth, free from any imperfections, burrs, or defects of any nature which may adversely affect their serviceability. The tooth face width of the gear teeth shall be not less than 3/8 inch with a maximum backlash of not more than 0.020 inch. The driving gear shall be malleable or gray iron with means provided for attaching the crank. The driving pinion or pinions shall be of steel. The gears shall run freely when assembled without evidence of binding or sticking.

3.5.3.1 *Gear ratio.* The gear ratio shall be not less than 1 to 1 for the low speed, and not less than 2.8 to 1 for the high speed.

3.5.4 *Speed-changing mechanism.* The speed-changing mechanism shall provide for a direct drive of the chuck in both the low- and high-speed operations. A manually operated knurled button or equally suitable means shall be conveniently located for per-

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mitting the selection of either low or high speed.

3.5.5 *Chuck.* The chuck shall be of steel with hardened steel jaws capable of securely gripping the shanks of drills or bits by hand tightening only. The jaws shall show a Rockwell hardness value of not less than 50 nor more than 58 on the C scale, or the jaws may be case-hardened to show a hardness value of not less than 87.0 nor more than 91.0 on the Rockwell 15*N* scale (superficial). The chuck shall be threaded on the spindle and shall be designed to provide a means for opening and closing the chuck jaws. The gripping surfaces of these jaws shall hold all drills or bits within the rated capacity of the drill within 0.030 of an inch run out for three jaw chucks and within 0.045 of an inch run out for two jaw chucks. (see 4.3.2). The portion of the interior surface of the chuck against which the jaws bear shall be sufficiently hard to prevent the jaws from cutting into the sleeve or body. The chuck sleeve shall be suitably threaded internally to properly screw on the chuck body. The outside diameter of the chuck sleeve shall be neatly knurled for actuating jaw adjustment and shall be capable of positively releasing the drill bit by hand, regardless of the drill load.

3.5.6 *Crank arm.* The crank arm shall be of steel or malleable iron and, unless otherwise specified, the length shall be nonadjustable, (see 6.2). The arm of the crank, measured from the center of the pivot to the center of the handle, shall be not less than 4 nor more than 6 inches.

3.5.7 *Breastplate.* The breastplate shall be either malleable iron, gray iron, or steel. The attaching bearing surface shall have a smooth machine finish, free of burrs and sharp edges. Surfaces other than attaching bearing surface shall be smoothly finished, free from burrs, excessive roughness and sharp edges. Method of attachment shall permit adjustment of the breastplate about the longitudinal axis of the drill.

3.5.8 *Handles.* Handle assemblies shall be detachable. The wooden portion shall be made of hardwood smoothly finished to afford a comfortable handgripping surface. This portion shall be lacquered or varnished to prevent the wood from absorbing moisture. Handles shall be provided with a steel ferrule as a support to prevent the handles from splitting or breaking.

3.5.8.1 *Crank handle.* The wooden portion of the crank handle shall be made free to rotate on the (steel) shaft.

3.5.8.2 *Sidegrip handle.* The wooden portion of the sidegrip handle shall be fixed to, or mounted on a steel shaft not less than 5/16 inch in diameter. The steel shaft shall extend into the wooden portion at least three times its diameter. The opposite end of this shaft shall project from the handle and expose at least 5/16 inch length of threads. The threaded end of the shaft shall fit into a threaded hole provided in the frame which shall be located on the opposite side from the driving gear.

3.5.9 *Spirit level.* When specified, (see 6.2), the breast drills shall be provided with a spirit level. The spirit level shall be accurately mounted either in or on the frame in a plane parallel with the longitudinal axis of the drill. The spirit level shall be positioned to afford protection against breakage and shall be easily visible while in use. Drills may have a sturdy steel rod or tubing extension on the breast-plate end of the frame. For frames of this style, the spirit level may be mounted in the tubing or in a drilled hole in the steel rod. The depth of the view-slot shall not exceed 40 percent of the outside diameter of the tubing or rod where the view-slot is located.

3.6 Drive. Breast drills shall have a direct drive in both low and high speeds.

3.6.1 *Style E, universal, two-jaw chucks.* The universal two-jaw chucks shall be capable of firmly and centrally holding square taper-shank bitstock drills and round straight-shank drills with diameters from one-eighth to one-half inch, inclusive.

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3.6.2 *Style C, three-jaw chucks.* The three-jaw chucks shall be capable of firmly and centrally holding straight round-shank drill bits up to and including one-half inch diameters.

3.6.3 *Drills shall be similar to figure* 1.

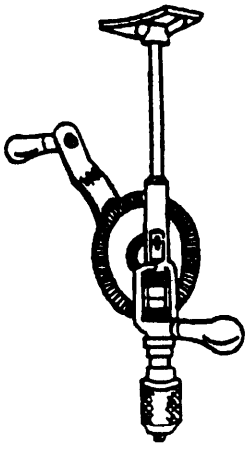


FIGURE 1. Drill, breast.

3.7 Finish and coating.

3.7.1 *Finish.* Breast drills shall be substantially free from pits, nodules, burrs, cracks or other detrimental defects.

3.8 Workmanship. Workmanship shall be first class in every respect and shall be free from any imperfections which may adversely affect the appearance, durability and serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2.2 *Inspection of the end item.* Inspection of the drills shall be performed as specified in the following paragraphs. A lot shall consist of drills of the same type and class including all spare parts. The sample unit shall be one completely fabricated and assembled drill.

4.2.2.1 *Visual examination.* Each sample unit shall be visually examined for the defects set forth in table I. The inspection level shall be level II, with an acceptable quality level (AQL) of 2.5 defects per hundred units.

TABLE I—*Classification of defects*

Examine	Defect
Finish	Not as specified.
	Breast plate not smoothly finished
	Detachable wooden handles not lacquered or varnished hardwood.
	Chuck jaw sleeve not knurled.
	Coating on driving gear, frame or breastplate not as specified.
	Crank not plated as required.
Design	Style of drill not as specified.
	Parts not interchangeable.
	Overall length of drill not within specified limits.
	Chuck fails to properly hold drills securely.
Material	Material of components not as specified.

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TABLE I—*Classification of defects (cont'd)*

Examine	Defect
Construction and Workmanship	Drill mechanism sticks or binds or has excessive looseness or backlash.
	Driving or pinion teeth do not mesh properly.
	Breastplate not adjustable.
Construction and Workmanship	Spindle not properly assembled.
	Gear ratios not as specified.
	Gears not machine cut or uniformly formed.
	Driving or pinion teeth broken or missing.
	Chuck operation not smooth.
	Speed changing mechanism defective.
	Grip handles loose.
Marking	Missing, incomplete, illegible.

4.2.2.2 *Dimensional examination.* Each sample unit shall be examined for dimensions specified in applicable paragraphs. Any nonconformance with requirements shall constitute a defect. The inspection level shall be S-3 with an AQL of 4.0 defects per hundred units.

4.2.2.3 *Testing.* Each sample unit shall be tested as specified in paragraphs 4.3.1 through 4.3.3 as applicable. Any determination not within specified requirements shall constitute a defect. The inspection level shall be level S-2 with an AQL of 2.5 defects per hundred units.

4.2.3 *Inspection preparation for delivery.* Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with MIL-H-15424.

4.3 Test procedures.

4.3.1 *Drilling test.* Sample breast drills shall be tested in the following manner in both the low and high speeds. The breast drills shall be subjected to drilling a soft cold-rolled steel sheet or plate in a vertical position. The steel sheet or plate shall be not less than 3/16 inch nor more than 1/4 inch thick. A maximum size drill bit within the capacity range of the breast drill shall be properly inserted in the chuck jaws and tightened by hand. The crank shall be turned with a forward stroke by hand at a speed not greater than 100 r.p.m. with a 55-pound pressure applied on the breastplate. The longitudinal axis of the drill shall be held at approximately right angles to the steel sheet or plate. When the drill is about to pierce the opposite side of the plate, the 55-pound load shall be released and normal pressure applied on the breastplate until the drill has pierced entirely through the plate. Each individual breast drill subjected to the test shall make at least two such drilling operations to determine that the breast drills are in compliance with the requirements of this specification. The breast drills and their component members shall show no mechanical defects and shall not fail, fracture, take permanent set or show any evidence of binding or undue wear.

4.3.2 *Concentricity test.* The frame of the sample drills shall be gripped in a bench vise, or by other suitable means, with the chuck in a horizontal plane. A test rod of any diameter within the capacity range of the chuck to be tested, shall be inserted in the drill chuck, and the chuck or drill sleeve tightened by hand. The test rod shall project at least 3.0 inches from the jaws when inserted to the full depth of the drill chuck. This test rod shall not run out more than 0.030 inch for three-jaw (style C) chucks, nor more than 0.045 inch for two-jaw (style B) chucks, total indicator reading in 360 degrees. At least the two extremities on the exposed length of the test rod shall be checked to insure that these concentricity requirements are met.

4.3.3 *Hardness test for jaws.* Jaws of sample drills hardened and tempered, shall be tested on a Rockwell "C" scale or, when case-hardened on a Rockwell "15N" scale (superficial) to determine conformance with this specification, (see 3.5.5). The Rockwell hardness test shall be conducted in accordance with method 243 of Fed. Test Method Std. 151.

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5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. 102 for definitions and applications of the various levels of preservation, packaging and packing.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified, (see 6.2).

5.1.1 *Level A.* Cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* Preservation and packaging which will afford adequate protection against corrosion, deterioration, and damage during shipment from supply source to the first receiving activity for immediate use, (vendor to user).

5.2 Packing. Packing shall be level A, B, or C, as specified, (see 6.2).

5.2.1 *Level A.* Drills, preserved and packaged as specified in 5.1 shall be packed in accordance with level A requirements of MIL-H-15424.

5.2.2 *Level B.* Drills, preserved and packaged as specified in 5.1 shall be packed in accordance with level B requirements of MIL-H-15424.

5.2.3 *Level C.* Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking.

5.3.1 *Military agencies.* In addition to any special marking required by the contract or order or herein, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The hand-operated breast drills covered by this specification are generally used for drilling holes in various kinds of metals, woods, plastics or other applicable materials.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Style of breast drill required (see 1.2).
(*c*) If idler gear is required (see 3.5.2).
(*d*) If adjustable length of crank arm is required (see 3.5.6).
(*e*) If spirit level is required (see 3.5.9).
(*f*) Applicable levels of packaging and packing required (see 5.1 and 5.2).
(*g*) Marking, if different (see 5.3).
(*h*) Repair parts, if required (see 3.3).

6.3 Transportation description. The proper transportation description is as follows:

Drills, hand, not otherwise indexed by name.
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

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